MAYER-VIETORIS LONG EXACT SEQUENCE FOR HOM AND EXT GROUPS ON A UNION OF TWO OPENS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and displays the adjunction identity that identifies the terms; no step is referred to the reader.

Editorial note. The counted spans carry 3 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-cohomology/retained-preamble.tex lines 1-81), a verbatim excerpt of the native preamble, and its Xy-pic setup renders the diagram of the statement.

Editorial note. The label lemma-mayer-vietoris-hom is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Mayer-Vietoris distinguished triangle for extension by zero. Same chapter, Lemma lemma-exact-sequence-lower-shriek, invoked at proof lines 8187-8188. Supplies the triangle to which the cohomological functor is applied.
- A distinguished triangle yields a long exact sequence of Hom groups. Derived Categories, Lemma derived-lemma-representable-homological, invoked at proof lines 8189-8190. Supplies the long exact sequence.
- Adjunction between extension by zero along an open immersion and restriction. Same chapter, Lemma lemma-adjoint-lower-shriek-restrict, invoked at proof lines 8191-8196. Supplies the identification of Hom groups upstairs with Hom groups on the open.

Source reference. Original lemma 0149: sources/stacks-chow/tags.txt, source lines 1464–1464. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 08BT: sources/stacks-cohomology/pinned-original-source.tex, source lines 8038–8038. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 08BU: sources/stacks-cohomology/pinned-original-source.tex, source lines 8084–8084. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $X = U \cup V$ be the union of two open subspaces of X. For objects E, F of $D(\mathcal{O}_X)$ we have a Mayer-Vietoris sequence*

$$\ldots \longrightarrow \mathrm{Ext}^{-1}(E_{U\cap V}, F_{U\cap V}) \longrightarrow \mathrm{Hom}(E, F) \longrightarrow \mathrm{Hom}(E_U, F_U) \oplus \mathrm{Hom}(E_V, F_V) \longrightarrow \mathrm{Hom}(E_{U\cap V}, F_{U\cap V})$$

where the subscripts denote restrictions to the relevant opens and the Hom*'s and* Ext*'s are taken in the relevant derived categories.*

Proof. Use the distinguished triangle of Lemma 08BU to obtain a long exact sequence of Hom's (from Derived Categories, Lemma 0149) and use that

$$\mathrm{Hom}_{D(\mathcal{O}_X)}(j_{U!}E|_U, F) = \mathrm{Hom}_{D(\mathcal{O}_U)}(E|_U, F|_U)$$

by Lemma 08BT. □